

[DC] XR Visualizations for Exploring 3D Dynamics

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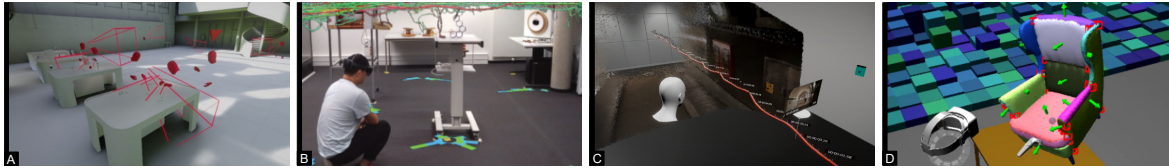


Figure 1: XR visualizations for exploring captured 3D dynamics: (A) In situ **monitoring** of dynamic 3D virtual museum visiting activity in MR. (B) Augmented reality lenses for human movement data **analysis**. (C) MR visualizations of tracked RGB-D videos for dynamic scene editing and **authoring**. (D) VR visualizations of static 3D scans with extracted geometry information to potentially help further **analyze** 3D dynamics.

ABSTRACT

Dynamic 3D information is all around us in everyday life. When digitized, dynamic 3D data find extensive application in domains such as education, archaeology, immersive journalism, VR therapy, and art. For example, analyzing 3D human motions and movements can offer valuable insights into human behavior and movement patterns. With the growing affordability of tracking and capturing devices, acquiring 3D data has become more and more accessible. At the same time, recent advances in VR and AR headsets make it possible for users to immerse themselves in the computer-generated virtual environment (VE), which is perfectly suited for exploring captured complex dynamic 3D datasets due to their inherent complexity and three-dimensional nature. However, since mixed reality technology is just emerging, current systems are not yet fully prepared to harness all its benefits for comprehensive dynamic 3D data exploration. In my dissertation project, I aim to explore the creative potential of **XR visualizations** for exploring complex dynamic 3D datasets. My dissertation aims to enrich XR visualization designs and interactive prototypes to better support 3D data exploration tasks in mixed reality.

Index Terms: Immersive Visualization, Mixed Reality

1 INTRODUCTION AND BACKGROUND

With the growing affordability of tracking and capturing devices, acquiring 3D data has become more and more accessible. Dynamic 3D information, such as human and device movements, user interactions, bustling streets, and crowded canteens, can be readily captured using consumer-level devices. This capability allows for the recording of dynamic 3D scenes, producing dynamic 3D data as output. Such captured dynamic 3D data find extensive applications in domains such as virtual museum, VR therapy, education, and art.

Very often, the data must be processed before it can be utilized in domain-specific applications. Traditionally, the automated processing of such data demands complex algorithms, and no universally effective methods exist. When it comes to interactive approaches, exploring complex 3D data on traditional 2D displays is generally considered less intuitive. Recent advances in VR and AR headsets enable users to immerse themselves in the computer-generated 3D virtual environment (VE), which is perfectly suited for exploring

captured dynamic and static 3D datasets due to the inherent complexity and three-dimensional nature of the data. Emerging mixed reality (MR) technologies offer the potential of immersively interfacing dynamic 3D data by displaying them in virtual space. The use of infinite virtual space allows for extensive data display while a broad field of view enhances observation capabilities. Besides, cutting-edge MR devices allow users to adjust through the whole reality-virtuality continuum, offering great flexibility in contextualizing the data exploration. Moreover, post-WIMP widgets can be developed to support the engaged immersive experience and customized for the individual needs of the applications. Additionally, incorporating multiple users, whether remote or co-located, to co-exist within a shared immersive environment facilitates collaborative work and sense-making.

When attempting to understand captured dynamic 3D data during post-processing, such as **during data analysis**, the raw data is often massive in size and complex spatially. A certain level of simplification and aggregation is desired to help users better understand it. For instance, traced 3D human movement data will cause a "spaghetti" issue when rendered directly in space. Tracked RGB-D videos are highly dynamic, the constant movement over time makes the observation difficult. Moreover, static 3D surroundings can influence 3D dynamics, and incorporating this context can further enhance the understanding of dynamic data. When analyzing 3D dynamics, XR visualizations should be designed to help. There are existing approaches proposing immersive XR visualizations for aggregating the movement data immersively on surrounding surfaces, physical referents [2], and human avatars.

Besides analyzing, to further explore the usage of dynamic 3D data for content **creation**, there is a need for tools to apply changes to the data for scene curation and manipulating the data in space and time. These tools require XR visualizations that enable users to immersively and intuitively adjust, modify, and interact with the data, enhancing their ability to develop rich, intricate, and interactive virtual environments.

When working with live 3D dynamic data, such as **monitoring** 3D dynamic information about visitors to support multiuser mixed reality experiences in large architectural spaces, challenges arise from the highly dynamic nature and varying scales of the data. These challenges necessitate carefully designed solutions including adaptive and in situ XR visualizations.

In summary, there is exciting potential for XR visualizations to support complex dynamic 3D data exploration. Still, challenges remain in designing and prototyping interactive XR visualizations to facilitate **data analysis**, **content authoring**, and **live monitoring** of such dynamic 3D data.

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2 RESEARCH QUESTIONS

In general, I aim to investigate the creative use of XR visualizations for a more engaged and immersive exploration of real-world captured dynamic 3D data. To address the challenges outlined previously, I have formulated my research questions as follows:

RQ1: How to design and prototype XR visualizations to help users monitor live dynamic 3D information in real-time?

RQ2: How to design and prototype XR visualizations to help users analyze recorded 3D dynamic information?

RQ3: How to design and prototype XR visualizations to support capture and recreate dynamic 3D scenes for content authoring?

3 RESEARCH TO DATE AND NEXT STEPS

In the following, I am going to detail how my published work addresses the research questions respectively.

3D human movement data offers valuable insights into the spatial dynamics of humans. In Pearl [2], together with my co-authors, I explored the record and immersive analysis of 3D human movement data to address RQ2. For this, we first prepared a simulated exhibition room ($8.6 \times 5.9\text{m} = 50.74\text{ m}^2$) with six installations of various shapes and heights. We invited nine university students to visit the room and recorded 27 minutes of spatial movement data with Oculus Quest 2 VR headsets. We propose to visualize the traced 3D human movement raw data with spatial trajectories, with arrows indicating the directions. The data is aligned with the room after a calibration step with a QR code. Further, we explored the use of physical surroundings to help filter (simplify) the visualization by introducing AR cubic-shaped lenses for locating regions of interest and examining spatial relations between lenses and movement trajectories. Additionally, we explored the use of AR lenses for aggregating (simplifying) the complex 3D movement data. We proposed methods for computing and visualizing movement patterns around and between individual AR lenses.

Besides [2], I further researched XR visualizations for exploring spatially tracked RGB-D video in mixed reality to address RQ3. Such video can be captured by a mobile RGB-D camera and has the potential to record dynamic scenes in public open places such as markets, canteens, or busy streets. The video is considered as 3D+Time data and can be used to recreate vivid immersive renderings in immersive environments. It can benefit research fields such as VR education, exposure therapy, immersive journalism, etc. With Dynascope [4], we captured such 3D data using an iPhone equipped with a LiDAR sensor while walking through a busy street and a crowded canteen. To support effective viewing of such dynamic data in immersive space, we extracted and rendered (displayed) the camera movement trajectory and used it as a major interface for manipulation. Time-varying information such as movement speed, audio, and number of people present in the video frame are designed to be shown alongside the camera trajectory. Additionally, we designed and implemented various aggregation (simplifying) strategies for the video content along the camera trajectories to enhance visual analysis. We proposed point-based “heatmaps” to show the density of points belonging to humans, indicating areas of crowding. Further, we proposed immersive spatial widgets as XR representations of key points in the original data for users to manipulate the data in space and time (as shown in Figure 1 (C)).

3D point cloud data can serve as a powerful addition for documenting the static contextual environment in which 3D dynamics take place. However, raw captured data can seldom be used without some sort of pre-processing. They often have to be classified into various parts in terms of geometry or semantics. To further address RQ2, in [1], we adopted and enhanced the continuous

level-of-detail rendering algorithm by introducing chunks. We implemented three “selection primitives” (sphere, cube, and clipping plane) and a selection panel for painting (manipulating) labels onto points. Further, in [3], we developed rendering strategies (interactive subsampling, magic lenses) for previewing point cloud data in great detail and enhanced animated XR visualizations to help inspect (display) the extracted structural information.

For the next step, in addition to the previous datasets, I would like to further investigate the creative potential of XR visualizations in supporting live 3D user activity data to help address RQ1. I would explore strategies to visualize human engagement with immersive visualizations **during** multi-user mixed reality experiences. That would assist hosts in understanding visitor activity and engagement and help them optimize the experiences accordingly. At the same time, I would explore the strategies to simplify and aggregate these visualizations across space and time.

4 DISSERTATION STATUS

I am a Ph.D. student in the Immersive Experience Lab at the Faculty of Computer Science, Technische Universität Dresden, supervised by Professor Matthew McGinity and Professor Raimund Dachsel. So far, I have published four papers on this research direction including two first-authored full papers [4, 3]. I aim to finish my doctoral study with two more peer-reviewed papers and a dissertation before January 2027.

5 QUESTIONS TO DISCUSS DURING DOCTORAL CONSORTIUM

Throughout my research, I utilize sketches and prototypical implementations to support my concepts. However, I often struggle with determining the most suitable method for the figures. While sketches offer clarity, they may show less labor for certain reviewers. How can I effectively convey my workload while maintaining clarity in my figures? Moreover, user studies are always asked by reviewers but might be less meaningful for my research directions. How should I deal with this situation?

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